NUMERICAL SEMIGROUPS OF DEPTH TWO WITH PRESCRIBED BUCHWEITZ PROFILES

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Abstract. For a finite set A of integers and $n \geq 1$ put $\beta_A(n) = |nA| - (2n-1)(|A|-1)$, where nA is the n-fold sumset of A. By a criterion of Buchweitz, the gapset G of a Weierstrass semigroup of genus at least two satisfies $\beta_G(n) \leq 0$ for every $n \geq 2$, and Eliahou, García-García, Marín-Aragón and Vigneron-Tenorio proved that the Buchweitz set $\{n \geq 2 : \beta_G(n) > 0\}$ of a numerical semigroup of genus at least two is finite. We show that the values $\beta_G(2), \dots, \beta_G(N)$ can be prescribed arbitrarily. For every $N \geq 2$ and all integers $a_2, \dots, a_N$ there is a numerical semigroup whose conductor is twice its multiplicity and whose gapset G satisfies $\beta_G(n) = a_n$ for $2 \leq n \leq N$ and $\beta_G(n) < 0$ for every $n > N$; its multiplicity and genus are explicit polynomials in N and $L = \max\{1, |a_2|, \dots, |a_N|\}$. Consequently, every finite subset of $\{2, 3, \dots\}$ is the Buchweitz set of a numerical semigroup. This answers three questions of Eliahou et al.: Buchweitz sets of numerical semigroups need not be intervals, the function β_G need not be unimodal, and every finite set occurs. We also give an example of genus 172 whose Buchweitz set is $\{3, 6\}$.

1. Introduction

Write $\mathbb{N}_0 = \{0, 1, 2, \dots\}$. For integers $a \leq b$ let $[a, b] = \{a, a+1, \dots, b\}$, and let $[a, \infty) = \{a, a+1, \dots\}$. A *numerical semigroup* is a submonoid S of $(\mathbb{N}_0, +)$ with finite complement. Its *gapset* is $G = \mathbb{N}_0 \setminus S$, its *genus* is $g = |G|$, its *multiplicity* is the least positive element m of S, its *conductor* is the least integer c with $[c, \infty) \subseteq S$, and its *depth* is $\lceil c/m \rceil$. We refer to [13] for background.

Every point P of a compact Riemann surface of genus g determines a numerical semigroup of genus g, the *Weierstrass semigroup* at P, which consists of the pole orders at P of the meromorphic functions that are holomorphic away from P. A numerical semigroup is called *Weierstrass* if it arises in this way. Hurwitz [7] asked which numerical semigroups are Weierstrass. Buchweitz [1] found a combinatorial obstruction: if S is a Weierstrass semigroup of genus $g \geq 2$ and G is its gapset, then the Riemann–Roch theorem gives

$$|nG| \leq (2n-1)(g-1) \qquad \text{for every } n \geq 2,$$

where nG denotes the set of sums of n elements of G, repetitions allowed; see also [2], [8, Proposition 2] and [6, Section 1]. The restriction $g \geq 2$ is needed: the only numerical semigroup of genus 1 is $\{0, 2, 3, \dots\}$, which is Weierstrass, and its gapset $G = \{1\}$ satisfies $|nG| = 1$ for every n. Buchweitz exhibited a semigroup of genus 16 violating the inequality for $n = 2$, and so showed that not every numerical semigroup is Weierstrass. Among the numerical semigroups of genus g, those violating the inequality for some n form a proportion that tends to 0 as $g \to \infty$ [8, Theorem 5], and explicit families of them have been constructed in [9, 6, 3].

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The following notation is due to Eliahou, García-García, Marín-Aragón and Vigneron-Tenorio [3, Definition 1.1, Notation 1.2]. For a finite set A of integers and $n \geq 1$ put

$$\beta_A(n) = |nA| - (2n-1)(|A|-1), \qquad \mathcal{B}(A) = \{n \geq 2 : \beta_A(n) > 0\}.$$

For a numerical semigroup S with gapset G, the *Buchweitz set* of S is $\mathrm{Buch}(S) = \mathcal{B}(G)$. Since $\beta_A(1) = 1$ for every finite A, the information carried by β_G lies in its values at $n \geq 2$, and we call the sequence $(\beta_G(n))_{n\geq 2}$ the *Buchweitz profile* of S. Buchweitz's criterion states that $\mathrm{Buch}(S) = \emptyset$ for every Weierstrass semigroup S of genus at least two.

Eliahou et al. proved that $\mathrm{Buch}(S)$ is finite for every numerical semigroup S of genus at least two [3, Theorem 3.3]. They also constructed numerical semigroups of depth two whose Buchweitz sets are prescribed intervals, namely every finite interval I with $|I| \geq 2$ and $\min I \in [2, 6]$, and the intervals $[7 + 2k, 7 + 4k]$ for $k \geq 1$ [3, Sections 3.2 and 3.3]. They then asked the following questions [3, Section 4].

Question 1.1 ([3, Question 4.1])**.** Let $A \subset \mathbb{Z}$ be a finite set, or more specifically a gapset. Is $\mathcal{B}(A)$ always an interval of integers?

Question 1.2 ([3, Question 4.2])**.** Even more so, is the function β_A unimodal?

Question 1.3 ([3, Question 4.3])**.** Let $T \subset \{2, 3, \dots\}$ be a finite set. Is there a finite set $A \subset \mathbb{N}_0$, or more specifically a gapset, with $\mathcal{B}(A) = T$?

Our main result shows that the Buchweitz profile of a numerical semigroup can begin with any prescribed finite sequence of integers, while all subsequent values are negative.

Theorem 1.4. *Let $N \geq 2$ and $L \geq 1$ be integers, and let $a_2, \dots, a_N$ be integers with $|a_n| \leq L$ for $2 \leq n \leq N$. Put $x = N + 1$ and*

$$m = 2x(4x+3)(x^2 + 2Lx + 1) + 1, \tag{1.1}$$

$$g = 8x^4 + (16L+6)x^3 + (12L+9)x^2 + (2L+6)x + 4. \tag{1.2}$$

There is a numerical semigroup S of multiplicity m, conductor $2m$ and genus g whose gapset G satisfies

$$\beta_G(n) = a_n \quad (2 \leq n \leq N) \qquad \text{and} \qquad \beta_G(n) < 0 \quad (n > N).$$

The multiplicity and the genus depend only on N and L, not on the sequence (a_n). Given arbitrary integers $a_2, \dots, a_N$, one may take $L = \max\{1, |a_2|, \dots, |a_N|\}$; in particular, the zero sequence is covered with $L = 1$. The number g in (1.2) is an upper bound for the least genus of a numerical semigroup realizing a given sequence. Taking $L = 1$ and $a_n \in \{0, 1\}$ we obtain the following consequence.

Corollary 1.5. *For every finite set $T \subseteq \{2, 3, \dots\}$ there is a numerical semigroup S of depth two with $\mathrm{Buch}(S) = T$. If $T \neq \emptyset$, then S is not a Weierstrass semigroup.*

In particular, Question 1.1 has a negative answer, since $\{2, 4\}$ occurs as a Buchweitz set, and Question 1.3 has a positive answer, also in its form for gapsets. Question 1.2 has a negative answer as well; see Remark 5.2. The semigroups of Theorem 1.4 have large genus. In Section 6 we give a numerical semigroup of multiplicity 169 and genus 172 with

$$(\beta_G(n))_{1\leq n\leq 7} = (1, 0, 1, 0, 0, 1, -5), \qquad \mathrm{Buch}(S) = \{3, 6\},$$

which answers the first two questions negatively with a semigroup of comparatively small genus.

Outline of the proof. The argument has two parts. Let $m \geq 3$ and let $C \subseteq [0, m-2]$ contain 0 and a positive element. Then

$$G_C = [1, m-1] \cup \{2m-1-c : c \in C\}$$

is the gapset of a numerical semigroup S_C of multiplicity m and conductor $2m$, and we show in Section 2 that

$$\beta_{G_C}(n) = |nC \cap [0, m-1]| - (2n-1)(|C|-1) \qquad \text{for every } n \geq 1. \tag{1.3}$$

Thus the Buchweitz profile of S_C is determined by the number of n-fold sums of C below m. In the second part (Sections 3 and 4) we choose C to consist of 0, three small elements $u, u+M, u+2M$, and h large elements $v_1, \ldots, v_h$ that lie in distinct nonzero residue classes modulo M. Every sum below m then contains at most one large element, and the number of sums below m that contain v_j depends only on a pair (d_j, t_j) of small integers. This gives an exact formula for $|nC \cap [0, m-1]|$ valid for all $n \geq 1$. The pairs are obtained from the prescribed sequence (a_n) by Euclidean division, and the same formula shows that $\beta_{G_C}(n)$ is negative for every $n > N$.

Relation to other work. The passage from G to the reflected set $(2m-1)-G$ for semigroups of depth two is used in [3, Section 3], and the semigroups S_C are those studied, under additional hypotheses, in [6, Section 4], where 2-Buchweitz semigroups of this shape are characterized by the size of a 2-fold sumset [6, Theorem 4.1]; see Remark 2.4. Identity (1.3) extends the computations made there to all n.

Nathanson [11] posed problems on oscillations in the sign of $|kA|-|kB|$ for finite sets $A, B \subseteq \mathbb{Z}$. Kravitz [10, Theorem 1.1] showed that, in every sufficiently large abelian group, the relative order of $|kA_1|, \ldots, |kA_d|$ can be prescribed independently for each $1 \leq k \leq s$; in the integers he also allowed prescribed equalities among these sizes, together with a prescribed relative order valid for all $k > s$ [10, Theorem 1.2]. Fox, Kravitz and Zhang [5, Theorem 2] showed that for all integers $c_1, \ldots, c_s$ there are finite sets $A, B \subseteq \mathbb{Z}$ with $|kA| - |kB| = c_k$ for $1 \leq k \leq s$, with explicit bounds on their size and diameter; their proof expresses these differences as linear combinations of auxiliary parameters, and the proof of Theorem 1.4 uses a similar device. These results compare several freely chosen sets with one another. In Theorem 1.4, by contrast, a single gapset G is compared with the quantity $(2n-1)(|G|-1)$ determined by G itself, and the sign of $\beta_G(n)$ is controlled for all $n > N$. Eliahou and Mazumdar [4, Theorem 4.12] showed that every sequence $(d_k)_{k \geq 0}$ of positive integers with $d_0 = 1$ satisfying Macaulay's growth conditions occurs as $(|kA|)_{k \geq 0}$ for a finite subset A of some abelian semigroup, and they note that the corresponding problem for abelian groups remains open [4, Section 5]. The construction below takes place in $\mathbb{N}_0$.

Organization. Section 2 proves identity (1.3). Section 3 defines the set C attached to the prescribed sequence, Section 4 counts its sums below m, and Section 5 proves Theorem 1.4 and Corollary 1.5. Section 6 contains the example of genus 172, and Section 7 collects some remarks.

2. Reflection for semigroups of depth two

For a finite set $A \subseteq \mathbb{Z}$ and $n \geq 1$, nA is the set of sums of n elements of A, repetitions allowed, and we put $0A = \{0\}$. For an integer y we write $y - A = \{y - a : a \in A\}$ and $y + A = \{y + a : a \in A\}$. Since $n(y - A) = ny - nA$, we have $|n(y - A)| = |nA|$ and hence $\beta_{y-A} = \beta_A$.

Lemma 2.1. *Let $m \geq 3$ and let $C \subseteq [0, m-2]$ with $0 \in C$. Then*

$$G_C = [1, m-1] \cup (2m-1-C)$$

is the gapset of a numerical semigroup S_C of multiplicity m, conductor $2m$ and genus $m-1+|C|$. Moreover $2m-1-G_C = C \cup [m, 2m-2]$.

Proof. Since $C \subseteq [0, m-2]$, the set $2m-1-C$ is contained in $[m+1, 2m-1]$ and contains $2m-1$; in particular it is disjoint from $[1, m-1]$, so $|G_C| = m-1+|C|$. Let $S_C = \mathbb{N}_0 \setminus G_C$. Every positive element of S_C is at least m, so the sum of two positive elements of S_C is at least $2m$, and every integer at least $2m$ lies in S_C. Hence S_C is a numerical semigroup. It contains m but no element of $[1, m-1]$, and it contains $[2m, \infty)$ but not $2m-1$, so its multiplicity is m and its conductor is $2m$. The last assertion follows from $2m-1-[1, m-1] = [m, 2m-2]$. □

Lemma 2.2. *Let $m \geq 3$ and let $C \subseteq [0, m-2]$ with $0 \in C$ and $\gamma = \max C > 0$. Put $A = C \cup [m, 2m-2]$. Then*

$$nA = nC \cup [m, 2n(m-1)] \qquad \text{for every } n \geq 1.$$

Proof. Put $J = [m, 2m-2]$. Sorting the summands of an element of nA according to how many of them lie in J, we obtain

$$nA = nC \cup \bigcup_{r=1}^{n} \big(rJ + (n-r)C\big), \qquad rJ = [rm, 2r(m-1)].$$

Fix $1 \leq r \leq n$ and put $D = (n-r)C$. The set D contains $0, \gamma, 2\gamma, \dots, (n-r)\gamma$ and is contained in $[0, (n-r)\gamma]$, so consecutive elements of D differ by at most γ. If $d < d'$ are consecutive elements of D, then

$$\min(rJ + d') = rm + d' \leq rm + d + \gamma \leq 2r(m-1) + d + 1 = \max(rJ + d) + 1,$$

because $\gamma \leq m-2 \leq r(m-2)$. Hence the translates $rJ + d$, $d \in D$, fill an interval, and

$$rJ + (n-r)C = I_r := [rm,\ 2r(m-1) + (n-r)\gamma].$$

For $1 \leq r \leq n-1$ we have $(n-r)\gamma \geq 1$, and therefore

$$\max I_r + 1 - \min I_{r+1} = (r-1)m - 2r + 1 + (n-r)\gamma \geq (r-1)(m-2) \geq 0.$$

Moreover $\min I_r < \min I_{r+1}$ and $\max I_r < \max I_{r+1}$, the latter because $2(m-1) > \gamma$. It follows that $I_1 \cup \dots \cup I_n$ is the interval from $\min I_1 = m$ to $\max I_n = 2n(m-1)$, which proves the lemma. □

For a finite set $C \subseteq \mathbb{N}_0$, an integer m and $n \geq 1$ we put

$$H_n(C) = |nC \cap [0, m-1]|,$$

the dependence on m being understood from the context.

Proposition 2.3. *Let $m \geq 3$ and let $C \subseteq [0, m-2]$ with $0 \in C$ and $\max C > 0$. Then the gapset G_C of Lemma 2.1 satisfies*

$$\beta_{G_C}(n) = H_n(C) - (2n-1)(|C|-1) \qquad \text{for every } n \geq 1.$$

Proof. Put $A = C \cup [m, 2m-2] = 2m-1-G_C$, so that $\beta_{G_C} = \beta_A$ and $|A| = |C| + m - 1$. Since $nC \subseteq [0, n(m-2)]$ and $n(m-2) < 2n(m-1)$, Lemma 2.2 gives the disjoint decomposition

$$nA = \big(nC \cap [0, m-1]\big) \sqcup [m, 2n(m-1)],$$

so $|nA| = H_n(C) + (2n-1)(m-1)$. Subtracting $(2n-1)(|A|-1) = (2n-1)(|C|+m-2)$ gives the claim. □

Remark 2.4. Proposition 2.3 reduces the Buchweitz profile of S_C to the counts $H_n(C)$. If $2\max C \le m-1$, then $H_2(C) = |2C|$ and $\beta_{G_C}(2) = |2C| - 3(|C|-1)$. The semigroups of [6, Theorem 4.1] have gapsets $[1, g-t] \cup \{2m-\alpha_1, \dots, 2m-\alpha_t\}$ with $m = g-t+1$, $t \ge 3$, $\alpha_1 > \cdots > \alpha_t = 1$ and $g \ge 2\alpha_1 + t - 1$. These are the sets G_C with $C = \{\alpha_1 - 1, \dots, \alpha_t - 1\}$, and the hypothesis on g is equivalent to $2\max C \le m-2$. That theorem shows that these semigroups are PF-semigroups and that they are 2-Buchweitz exactly when $|2\{\alpha_1, \dots, \alpha_t\}| > 3(t-1)$; for them the case $n = 2$ of Proposition 2.3 recovers the count of $|2G_C|$ in its proof. Likewise, the decompositions underlying the sumset formulas in [3, Propositions 3.4 and 3.5] are instances of Lemma 2.2.

3. The construction

Throughout this section and the next we fix integers $N \ge 2$ and $L \ge 1$ and integers $a_2, \dots, a_N$ with $|a_n| \le L$. We also put $a_1 = 1$; note that $|a_1| \le L$. Define

$$\begin{aligned} x &= N+1, \qquad h = x^2 + 2Lx, \qquad K = 2x, \\ M &= h+1, \qquad u = (2K+1)M, \qquad m = K(u+2M)+1. \end{aligned} \tag{3.1}$$

Then $m = 2x(4x+3)M + 1$, which is (1.1). For $2 \le n \le N$ define

$$R_n = 2h + 5 - 2n + a_n - a_{n-1}, \qquad q_n = \left\lfloor \frac{R_n}{2n-1} \right\rfloor, \qquad r_n = R_n - (2n-1)q_n, \tag{3.2}$$

so that $0 \le r_n \le 2n-2$, and put $q_1 = h$.

Lemma 3.1. *We have $R_n > 0$ for $2 \le n \le N$, and*

$$h = q_1 > q_2 > \cdots > q_N \ge 0.$$

Proof. Since $|a_n - a_{n-1}| \le 2L$ and $n \le N = x - 1$,

$$R_n \ge 2h + 5 - 2N - 2L = 2x^2 + 4Lx + 7 - 2x - 2L > 0,$$

so every q_n is nonnegative. For $n = 2$ we have $R_2 = 2h + a_2 < 3h$, because $a_2 \le L < h$; hence $q_2 \le R_2/3 < h$. Now let $3 \le n \le N$. A direct computation gives

$$(2n-1)R_{n-1} - (2n-3)R_n = 4h + 8 + (4n-4)a_{n-1} - (2n-1)a_{n-2} - (2n-3)a_n.$$

Using $|a_k| \le L$ for $1 \le k \le N$ and $n \le x - 1$, the right-hand side is at least

$$4h + 8 - 8L(n-1) = 4x^2 + 8L(x-n+1) + 8 > 4x^2 > (2n-3)(2n-1).$$

Dividing by $(2n-3)(2n-1)$ we obtain $R_{n-1}/(2n-3) > R_n/(2n-1) + 1$. Since the floor function is nondecreasing and $\lfloor y+1 \rfloor = \lfloor y \rfloor + 1$, it follows that $q_{n-1} \ge q_n + 1$. □

By Lemma 3.1 the integers $e_n = q_{n-1} - q_n$, $2 \le n \le N$, are positive. We form a multiset P of pairs of integers as follows. For each $2 \le n \le N$, P contains the pair (n, r_n) once and the pair $(n, 0)$ with multiplicity $e_n - 1$; in addition, P contains the pair $(N+1, 0)$ with multiplicity q_N. The number of pairs in P is

$$\sum_{n=2}^{N} e_n + q_N = q_1 = h.$$

We enumerate P in an arbitrary order as $(d_1, t_1), \dots, (d_h, t_h)$. For every j we have

$$2 \le d_j \le N+1, \qquad 0 \le t_j \le 2d_j - 2, \qquad t_j = 0 \text{ if } d_j = N+1, \tag{3.3}$$

and for $2 \le n \le N$,

$$\big|\{j : d_j \ge n+1\}\big| = \sum_{k=n+1}^{N} e_k + q_N = q_n, \qquad \sum_{j:\, d_j = n} t_j = r_n. \tag{3.4}$$

Finally, for $1 \le j \le h$ we put

$$\rho_j = (d_j - 1)u + (t_j - 1)M + j, \qquad v_j = m - 1 - \rho_j, \tag{3.5}$$

and we define

$$U = \{u,\, u+M,\, u+2M\}, \qquad V = \{v_1, \dots, v_h\}, \qquad C = \{0\} \cup U \cup V. \tag{3.6}$$

4. Counting sums below m

We keep the notation of Section 3. Recall that $K = 2N + 2$ and $h = M - 1$.

Lemma 4.1. *For $1 \le j \le h$ we have*

$$0 < \rho_j \le N(u+2M) - 1, \qquad (N+2)(u+2M) < v_j \le m - 2, \qquad v_j \equiv -j \pmod{M}.$$

Moreover $u + 2M < m/2 < v_j$ for every j, and the elements of $\{0\} \cup U$ are divisible by M. Consequently $|C| = h + 4$, $C \subseteq [0, m-2]$, and every element of $nC \cap [0, m-1]$, $n \ge 1$, is a sum of n elements of C at most one of which lies in V.

Proof. Since $d_j \ge 2$, $t_j \ge 0$ and $j \ge 1$, we have $\rho_j \ge u - M + 1 > 0$. Since $d_j - 1 \le N$, $t_j - 1 \le 2d_j - 3 \le 2N - 1$ and $j \le h = M - 1$, we have $\rho_j \le Nu + (2N-1)M + M - 1 = N(u+2M) - 1$. As $m - 1 = K(u+2M)$ and $K - N = N + 2$, this gives $v_j \ge (N+2)(u+2M) + 1$, while $\rho_j \ge 1$ gives $v_j \le m - 2$. Since $m - 1$ and u are multiples of M, we get $v_j \equiv -\rho_j \equiv -j \pmod{M}$. Next,

$$\frac{m}{2} = (N+1)(u+2M) + \frac{1}{2},$$

so $u + 2M < m/2 < (N+2)(u+2M) < v_j$. The residues $-1, \dots, -h$ modulo $M = h + 1$ are distinct and nonzero, while 0, u, $u + M$ and $u + 2M$ are distinct multiples of M; hence $|C| = h + 4$. Finally, the sum of two elements of V exceeds m, which gives the last assertion. □

For $\ell \ge 0$ let Σ_ℓ be the set of sums of exactly ℓ elements of U, repetitions allowed, with $\Sigma_0 = \{0\}$. Writing an element of U as $u + iM$ with $i \in \{0, 1, 2\}$, we obtain

$$\Sigma_\ell = \{\ell u + bM : 0 \le b \le 2\ell\}, \qquad |\Sigma_\ell| = 2\ell + 1.$$

Lemma 4.2. *The sets $\Sigma_0, \Sigma_1, \dots, \Sigma_K$ are pairwise disjoint subsets of $[0, m-1]$, and $\min \Sigma_\ell \ge m$ for every $\ell \ge K + 1$.*

Proof. For $0 \le \ell \le K$ we have $\max \Sigma_\ell = \ell u + 2\ell M < \ell u + u = \min \Sigma_{\ell+1}$, because $2\ell M \le 2KM < u$. Thus $\Sigma_0, \dots, \Sigma_K$ lie in successive disjoint ranges, and $\max \Sigma_K = K(u+2M) = m - 1$. For $\ell \ge K + 1$ we have $\min \Sigma_\ell \ge (K+1)u = K(u+2M) + M = m - 1 + M \ge m$. □

For a statement $\mathcal{P}$ we write $\mathbf{1}[\mathcal{P}]$ for its indicator, equal to 1 if $\mathcal{P}$ holds and 0 otherwise.

Proposition 4.3. *For every $n \ge 1$,*

$$H_n(C) = \big(\min(n, K) + 1\big)^2 + \sum_{j=1}^{h} \Big(\big(\min(n-1, d_j - 2) + 1\big)^2 + t_j\, \mathbf{1}[n \ge d_j] \Big). \tag{4.1}$$

Proof. By Lemma 4.1, an element of $nC \cap [0, m-1]$ either lies in Σ_ℓ for some $0 \le \ell \le n$, or lies in $v_j + \Sigma_\ell$ for some j and some $0 \le \ell \le n - 1$. Conversely, since $0 \in C$, all elements of these sets lie in nC. Hence

$$nC \cap [0, m-1] = X_0 \cup X_1 \cup \cdots \cup X_h,$$

where

$$X_0 = \Big(\bigcup_{\ell=0}^{n} \Sigma_\ell \Big) \cap [0, m-1], \qquad X_j = \Big(\bigcup_{\ell=0}^{n-1} (v_j + \Sigma_\ell) \Big) \cap [0, m-1] \quad (1 \le j \le h).$$

Elements of X_0 are divisible by M, and elements of X_j are congruent to $-j$ modulo M by Lemma 4.1. So the sets $X_0, \dots, X_h$ are pairwise disjoint.

By Lemma 4.2, X_0 is the disjoint union of $\Sigma_0, \dots, \Sigma_{\min(n,K)}$, and therefore

$$|X_0| = \sum_{\ell=0}^{\min(n,K)} (2\ell+1) = \big(\min(n,K)+1\big)^2.$$

Now fix j. An element $v_j + \ell u + bM$ of $v_j + \Sigma_\ell$ lies in $[0, m-1]$ if and only if $\ell u + bM \le \rho_j$. We distinguish three ranges of ℓ.

If $\ell \le d_j - 2$, then all of $v_j + \Sigma_\ell$ lies below m. Indeed, $\max \Sigma_\ell \le (d_j-2)(u+2M)$, and by (3.3)

$$\rho_j - (d_j-2)(u+2M) = u + (t_j + 3 - 2d_j)M + j \ge u - (2N-1)M + 1 > 0.$$

If $\ell = d_j - 1$, then the condition $(d_j-1)u + bM \le \rho_j$ reads $b \le t_j - 1 + j/M$. Since $0 < j/M < 1$, it holds exactly for $b \in \{0, 1, \dots, t_j - 1\}$, and these values satisfy $b \le 2\ell$ because $t_j \le 2d_j - 2$. So exactly t_j elements of $v_j + \Sigma_{d_j-1}$ lie below m.

If $\ell \ge d_j$, then no element of $v_j + \Sigma_\ell$ lies below m, because

$$d_j u - \rho_j = u - (t_j - 1)M - j \ge u - (2N-1)M - (M-1) > 0.$$

The sets Σ_ℓ with $\ell \le d_j - 1 \le N < K$ are pairwise disjoint by Lemma 4.2, and only $\ell \le n-1$ occurs in X_j. Hence

$$|X_j| = \sum_{\ell=0}^{\min(n-1, d_j-2)} (2\ell+1) + t_j\, \mathbf{1}[n-1 \ge d_j - 1] = \big(\min(n-1, d_j-2)+1\big)^2 + t_j\, \mathbf{1}[n \ge d_j].$$

Summing over j proves (4.1). □

5. Proof of the main results

Proof of Theorem 1.4. Let C be the set (3.6) and let $S = S_C$ be the numerical semigroup of Lemma 2.1, with gapset $G = G_C$. Note that $m \ge 3$. By Lemma 4.1, $C \subseteq [0, m-2]$, $0 \in C$, $\max C > 0$ and $|C| = h + 4$. Lemma 2.1 shows that S has multiplicity m, conductor $2m$ and genus $m - 1 + |C| = m + h + 3$. Substituting (3.1) gives

$$m + h + 3 = 2x(4x+3)(x^2 + 2Lx + 1) + x^2 + 2Lx + 4,$$

which expands to (1.2). Write $H_n = H_n(C)$. By Proposition 2.3,

$$\beta_G(n) = H_n - (2n-1)(h+3) \qquad (n \ge 1),$$

and so, for $n \ge 2$,

$$\beta_G(n) - \beta_G(n-1) = \Delta_n - 2(h+3), \qquad \Delta_n = H_n - H_{n-1}. \tag{5.1}$$

We compute Δ_n from Proposition 4.3. For $n \ge 2$ and $2 \le d \le N+1$,

$$\big(\min(n-1, d-2)+1\big)^2 - \big(\min(n-2, d-2)+1\big)^2 = (2n-1)\,\mathbf{1}[d \ge n+1],$$

and $\mathbf{1}[n \ge d] - \mathbf{1}[n-1 \ge d] = \mathbf{1}[n = d]$. Therefore

$$\Delta_n = (2n+1)\,\mathbf{1}[n \le K] + (2n-1)\big|\{j : d_j \ge n+1\}\big| + \sum_{j:\, d_j = n} t_j \qquad (n \ge 2). \tag{5.2}$$

The prescribed range. Let $2 \le n \le N$. Then $n \le K$, and (5.2) together with (3.4) and (3.2) gives

$$\Delta_n = 2n + 1 + (2n-1)q_n + r_n = 2n + 1 + R_n = 2(h+3) + a_n - a_{n-1}.$$

By (5.1), $\beta_G(n) - \beta_G(n-1) = a_n - a_{n-1}$ for $2 \le n \le N$. Since $\beta_G(1) = 1 = a_1$, induction gives $\beta_G(n) = a_n$ for $1 \le n \le N$.

The case $n = N+1$. By (3.3), no d_j exceeds $N+1$, and $t_j = 0$ whenever $d_j = N+1$. Since $N+1 \le K$, (5.2) gives $\Delta_{N+1} = 2N+3$, and therefore

$$\beta_G(N+1) = a_N + 2N + 3 - 2(h+3) \le L + 2N - 2h - 3 < 0,$$

because $h = x^2 + 2Lx > L + N$.

The range $N+2 \le n \le K$. Here (5.2) gives $\Delta_n = 2n+1 \le 2K+1 = 4x+1 < 2(h+3)$, so $\beta_G(n) < \beta_G(n-1)$ by (5.1).

The range $n > K$. Here $\Delta_n = 0$, so $\beta_G(n) = \beta_G(n-1) - 2(h+3)$.

Combining the last three steps, $\beta_G(n) < 0$ for every $n > N$. □

Remark 5.1. For $n \ge K$ the last step gives $\beta_G(n) = \beta_G(K) - 2(h+3)(n-K)$. This agrees with the proof of [3, Theorem 2.3], where [3, Corollary 2.2] yields $\beta_G(n) = (f - 2g + 1)n + b + 1 - g$ for all sufficiently large n, with f the Frobenius number and b an integer. Here $f = 2m - 1$ and $g = m + h + 3$, so $f - 2g + 1 = -2(h+3)$.

Proof of Corollary 1.5. Let $N = \max(T \cup \{2\})$, $L = 1$, and $a_n = \mathbf{1}[n \in T]$ for $2 \le n \le N$. The semigroup S given by Theorem 1.4 has conductor twice its multiplicity, hence depth two, and its gapset G satisfies $\beta_G(n) > 0$ exactly for $n \in T$ among $n \ge 2$. Thus $\mathrm{Buch}(S) = T$. If $T \ne \emptyset$, then S, whose genus is at least two, is not Weierstrass by Buchweitz's criterion. □

Remark 5.2. We call a sequence (x_n) of integers, indexed by an interval of integers, *unimodal* if there is an index k such that (x_n) is nondecreasing for $n \le k$ and nonincreasing for $n \ge k$. Theorem 1.4 with $N = 5$, $L = 1$ and $(a_2, a_3, a_4, a_5) = (0, 1, 0, 1)$ gives a numerical semigroup of depth two whose gapset satisfies $(\beta_G(n))_{1 \le n \le 5} = (1, 0, 1, 0, 1)$. Neither $(\beta_G(n))_{n \ge 1}$ nor $(\beta_G(n))_{n \ge 2}$ is unimodal, so Question 1.2 has a negative answer. Proposition 6.1 below gives an example of much smaller genus.

6. An example of small genus

The semigroups of Theorem 1.4 have large genus; for $N = 2$ and $L = 1$, Theorem 1.4 gives $m = 1441$ and $g = 1459$. The following example answers Questions 1.1 and 1.2 with a semigroup of much smaller genus. It is not obtained from the construction of Section 3, but its analysis follows the same pattern.

Proposition 6.1. *Let* $m = 169$ *and* $C = \{0, 26, 28, 97\}$, *and let* S *be the numerical semigroup with gapset*

$$G = G_C = [1, 168] \cup \{240, 309, 311, 337\}.$$

Then S *has multiplicity* 169, *conductor* 338 *and genus* 172. *Its gapset satisfies*

$$\beta_G(n) = \binom{\min(n,6)+2}{2} + \binom{\min(n-1,2)+2}{2} - 3(2n-1) \qquad (n \ge 1),$$

so that

n	1	2	3	4	5	6	7	8
$\beta_G(n)$	1	0	1	0	0	1	−5	−11

and $\beta_G(n) = 37 - 6n$ *for* $n \ge 6$. *In particular* $\mathrm{Buch}(S) = \{3, 6\}$, *and* S *is not a Weierstrass semigroup.*

Proof. Lemma 2.1 applies with $m = 169$, and $2m - 1 - C = \{337, 311, 309, 240\}$, which gives the multiplicity, the conductor and the genus $168 + 4 = 172$. By Proposition 2.3, $\beta_G(n) = H_n(C) - 3(2n-1)$, and it remains to count $nC \cap [0, 168]$.

The sums of exactly ℓ elements of $\{26, 28\}$ form the set $\Sigma_\ell = \{26\ell + 2b : 0 \le b \le \ell\}$ with $\ell + 1$ elements. For $\ell \le 6$ we have $\max \Sigma_\ell = 28\ell < 26(\ell + 1) = \min \Sigma_{\ell+1}$, and $\max \Sigma_6 = 168$, while $\min \Sigma_\ell \ge 182$ for $\ell \ge 7$. Since $2 \cdot 97 > 168$, an element of $nC \cap [0, 168]$ contains the summand 97 at most once. The elements of the sets Σ_ℓ are even, whereas $97 + \Sigma_\ell$ consists of odd numbers. Finally, $97 + 26\ell + 2b \le 168$ holds for every $0 \le b \le \ell$ when $\ell \le 2$, since $97 + 56 = 153$, and fails for $\ell \ge 3$, since $97 + 78 = 175$. Therefore

$$H_n(C) = \sum_{\ell=0}^{\min(n,6)} (\ell + 1) + \sum_{\ell=0}^{\min(n-1,2)} (\ell + 1) = \binom{\min(n, 6) + 2}{2} + \binom{\min(n - 1, 2) + 2}{2},$$

which gives the stated formula. For $n \ge 6$ we get $H_n(C) = 28 + 6 = 34$ and $\beta_G(n) = 37 - 6n$, which is negative for $n \ge 7$. The table follows by evaluation, and $\mathrm{Buch}(S) = \{3, 6\}$. The last assertion is Buchweitz's criterion. □

7. Concluding remarks

Remark 7.1. Theorem 1.4 prescribes finitely many values of the Buchweitz profile and only the sign of the remaining ones. The tail cannot be prescribed arbitrarily: if S has multiplicity at least 3, then β_G is eventually affine with slope $f - 2g + 1$, as recalled in Remark 5.1, and $f - 2g + 1 \le 0$ by [3, Proposition 3.2]. We do not address which infinite sequences occur as Buchweitz profiles.

Remark 7.2. The semigroups of Theorem 1.4 have depth two, and their embedding dimension is close to their multiplicity: every element of $S \cap [m, 2m - 1]$ is a minimal generator, and this set has $m - |C|$ elements. It would be interesting to know whether prescribed profiles can be realized by numerical semigroups of bounded embedding dimension, and to determine the least genus of a numerical semigroup with a given Buchweitz set T. For $T = \{3, 6\}$, Theorem 1.4 with $N = 6$ and $L = 1$ gives genus 27843, while Proposition 6.1 gives genus 172. For comparison, Komeda's computations show that there is no 2-Buchweitz semigroup of genus g with $2 \le g \le 15$; see [9, p. 161] and [6, Section 2].

Remark 7.3. When $T = \emptyset$, Buchweitz's criterion gives no information, and we do not determine whether the corresponding semigroups of Corollary 1.5 are Weierstrass. An empty Buchweitz set also occurs in a much simpler way: symmetric numerical semigroups of multiplicity at least 3 satisfy $\beta_G(n) = 0$ for all $n \ge 2$ [9, 12].

Use of AI-assisted tools

AI-assisted tools were used in the research and preparation of this manuscript.

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